

PROPOSAL DOCUMENT

VOSS Labs

Institutional Proposal

A proposal for the establishment of an officially recognised, student-run open source software laboratory at Vidyalankar Institute of Technology, Mumbai.

Submitted to: The Principal & Management, Vidyalankar Institute of Technology, Mumbai

Submitted by: VOSS Labs Founding Team

Date: May 2026

"Built by students. For students. In public."

Executive Summary

An overview of VOSS Labs, its mission, and the institutional partnership being sought.

VOSS Labs (Vidyalankar Open Source Software Labs) is a student-run open source software laboratory founded in April 2026. Its mission is to build real digital infrastructure for Vidyalankar Institute of Technology. The lab operates on a single principle: "No forms. No interviews. Just commits." Membership is earned through contribution. There are no elections, no appointed positions, and no gatekeeping. A student's GitHub pull requests are their resume.

In its first eight weeks of operation, VOSS Labs has launched three active open source projects, attracted nine contributors (seven from VIT and two external international contributors), and released all code under the MIT licence on GitHub. The flagship project, VERP, is a full ERP system purpose-built for the college, with twenty commits and seven contributors including two from outside India. vboard is an events management platform designed to replace ad-hoc Google Forms for campus events. vask is a privacy-first anonymous campus forum accessible entirely via SSH, with thirty-three commits, fourteen releases, and a current stable version of v1.1.2. The VOSS GitHub organisation is open and global — the lab has already attracted contributors from outside India who have verified and improved its code.

This proposal requests three things from the Institute administration. First, official recognition of VOSS Labs as a student technical society. Second, a faculty advisor to provide mentorship, academic guidance, and administrative liaison. Third, access to shared lab space for collaborative coding sessions. VOSS Labs places no financial burden on the Institute. All tools are open source, all infrastructure is self-funded through contributions, and the lab operates at zero cost to the college. As the lab grows, it will explore external grants and industry partnerships to sustain its operations.

The expected outcomes extend across multiple institutional dimensions. For accreditation, VOSS Labs provides documented evidence for NAAC Criterion III (Research, Innovations & Extension) and supports NBA Programme Outcome attainment through measurable student activity. For students, every contributor graduates with a public, verifiable portfolio of production code that recruiters can independently evaluate. For the college, the lab creates a direct pipeline to Google Summer of Code, where selected students earn stipends of \$1,500 to \$6,000.

"No forms. No interviews. Just commits." — The VOSS Labs ethos. Contribution is the only credential.

The Problem & The Opportunity

Three interrelated gaps — and how one intervention addresses all of them.

1. The Skills Gap

Students complete their coursework with strong theoretical foundations but limited exposure to real-world software development. Most have never collaborated on a multi-contributor codebase, never handled a production incident, and never shipped software to actual users. Industry employers increasingly value demonstrated open source contributions over certificates and coursework. VIT students currently lack a structured pathway to build these credentials while still in college.

2. The Infrastructure Gap

The college relies on manual administrative processes and fragmented digital tools. Attendance is tracked on paper. Campus events are managed through Google Forms and scattered WhatsApp messages. There is no unified digital infrastructure connecting students, faculty, and administration. The irony is that the students who experience these problems most intimately are also the ones best positioned to build solutions — if given the structure and support to do so.

3. The Opportunity

A student-run open source laboratory addresses both gaps simultaneously. Students build production-grade software and learn the full development lifecycle: version control, continuous integration, code review, automated testing, and deployment. The college receives MIT-licensed tools with no vendor lock-in and no licensing fees. Students graduate with verifiable public portfolios. The Institute gains innovation metrics for accreditation bodies. Every commit serves both the student's career and the college's infrastructure.

4. Peer Comparison

The following table compares VIT Mumbai's current standing against peer institutions that have established student-run open source labs:

Institution	Student Innovation Lab	Open Source Focus	GSoC Selections (2024)
IIT Bombay	Yes — WnCC, DevCom	Strong	40+
IIT Kharagpur	Yes — KOSS	Strong	25+
IIT Roorkee	Yes — SDS Labs	Strong	15+
MIT Manipal	Yes — FOSSMU	Dedicated	8+
RPI (USA)	Yes — RCOS	Strong	10+
VIT Mumbai	No formal lab	None currently	0

Current Projects & Impact

Active open source projects under development at VOSS Labs as of May 2026.

Project	Description	Technology Stack	Status	Contributors
VERP	Open-source ERP for VIT — student records, attendance, course management	Next.js 16 , TypeScript , PostgreSQL , Drizzle	Active — 20 commits, schema complete	7 (incl. 2 international)
vboard	Events platform — replaces Google Forms for campus event registration	TypeScript , Vite , Drizzle	Phase 0 shipped, UI on mock data	1
vask	Anonymous campus forum via SSH — no email, privacy-first	Go , SQLite	Production — 33 commits, 14 releases, v1.1.2	2

Project Details

VERP — Vidyalankar Enterprise Resource Planning

VERP is the flagship project of VOSS Labs: a comprehensive ERP system designed specifically for Vidyalankar Institute of Technology. Built on Next.js 16 with TypeScript and PostgreSQL, VERP handles student records, attendance tracking, and course management. The database schema is complete and production-ready. With twenty commits and seven contributors — including two international developers who found the project through GitHub and submitted verified improvements — VERP demonstrates that VOSS projects can attract global attention and contribution.

vboard — Campus Events Platform

vboard replaces the current ad-hoc system of Google Forms and WhatsApp announcements for campus event management. Built with TypeScript, Vite, and Drizzle ORM, Phase 0 has shipped with a complete UI running on mock data. The project is currently seeking additional contributors to connect the frontend to live data and build out the event registration flow.

vask — Anonymous SSH Forum

vask is a privacy-first anonymous campus forum with a novel interaction model: students access it entirely through SSH, with no email address required and no personal data collected. Written in Go with SQLite, vask has reached production stability with thirty-three commits, fourteen releases, and a current version of v1.1.2. Two contributors have built and maintained the entire system, demonstrating the depth of output a small focused team can achieve.

What Peer Labs Have Achieved

Student-run open source labs at peer institutions have demonstrated consistent, measurable impact over multiple years:

- IIT Kharagpur's KOSS runs Kharagpur Winter of Code (KWoC) annually — 1,393 students trained, 503 pull requests merged across participating projects.
- IIT Roorkee's SDSLabs builds tools the campus actually uses daily — their noticeboard and scheduling applications are institution-critical infrastructure.
- MIT's SIPB (Student Information Processing Board, founded 1969) runs campus-critical services including the Hyades cluster and scripts.mit.edu, with 216 lifetime members — students maintain infrastructure the college depends on.
- Rensselaer Polytechnic Institute's RCOS is a credited course with 160+ members per semester — projects live for years, not semesters, creating lasting institutional value.
- FOSS United now funds college FOSS clubs with grants up to Rs. 1 lakh, event sponsorships of Rs. 25,000, swags for core teams, and mentorship — even tier-3 colleges with active labs receive support.

Benefits to the Institution

VOSS Labs delivers value across every stakeholder group at Vidyalandkar Institute of Technology.

Stakeholder	Key Benefits
Principal / Director	NAAC Criterion III — Research, Innovations & Extension evidence. Innovation metrics, student publications, documented outcomes. Zero-cost infrastructure — MIT licence, GitHub, open source tooling. No capital expenditure.
HOD	NBA PO Attainment — PO5 (Modern Tools), PO3 (Design/Development), PO12 (Life-long Learning). Faculty-student collaboration on funded projects. Competition achievements and measurable outcomes.
TPO	GitHub-verifiable skill portfolios — every student graduates with public contribution history recruiters can independently verify. GSoC pipeline (\$1,500-\$6,000 per student). Direct company recruitment from open source activity.
Faculty	Research assistants for projects. Industry connections through open source ecosystem. GitHub Student Developer Pack (\$100,000+ value in free developer tools) available for all students.
Students	Real-world experience with production codebases. Public contribution history — a resume that cannot be faked. Mentorship and leadership development through peer code review and release management.

Accreditation Alignment — NBA Programme Outcomes

PO	Description	VOSS Labs Evidence
PO 3	Design / Development of Solutions	Students design complete systems (VERP, vboard, vask) addressing real campus problems
PO 5	Modern Tool Usage	Industry-standard tools: Git, GitHub Actions, TypeScript, Next.js, Go, Docker
PO 9	Individual & Team Work	Distributed teams, pull requests, code reviews, release coordination
PO 10	Communication	Technical documentation, issue reports, READMEs, commit messages
PO 12	Life-long Learning	Self-directed, continuous learning of new frameworks and technologies

Additional Government & Accreditation Benefits

- NAAC Criterion III (Research, Innovations & Extension) — student-led open source projects with public repositories provide documented, verifiable evidence of innovation activity that can be submitted during accreditation cycles.
- NBA Programme Outcomes — PO 3, 5, 9, 10, and 12 are all demonstrably met through sustained open source contribution, with public GitHub activity serving as auditable evidence.
- AICTE Innovation Cell encourages institution-level innovation labs — active open source labs qualify for recognition under the Innovation Cell framework.
- Institutions with documented student innovation activities are prioritised for AICTE-funded Quality Improvement Programmes (QIP) and Modernisation and Removal of Obsolescence (MODROBS) grants.
- MHRD's Innovation Cell and the Institution's Innovation Council (IIC) score student-led technical projects as Category A innovation activities — the highest classification for institutional innovation metrics.

Growth Trajectory

Metrics and milestones from founding through Year 1.

Metric	Current (8 Weeks)	Next Semester Target	Year 1 Target
Active Contributors	9	25	100+
Projects Maintained	3	4	6+
Total Commits	73+	200+	500+
Students with Merged PRs	5	20	50+
GitHub Stars (all projects)	15+	50+	100+
Releases Shipped	14	20	40+
GSoC Applications	0	2	5+
Collaborative Coding Sessions	4	10	15+

Growth Levers

Four factors will drive VOSS Labs' growth over the next academic year:

1. **Technical Representative Network:** Technical Representatives (TRs) from each academic year serve as the primary outreach channel. TRs are invited to participate in VOSS and form teams for college-facing projects. This peer-to-peer model scales organically as each year's TRs identify and recruit interested contributors.
2. **Good-First-Issue Curation:** Every VOSS repository maintains a curated set of beginner-friendly issues tagged "good first issue." New contributors can make their first pull request within hours of discovering the lab, not weeks. This dramatically reduces the friction of first-time contribution.
3. **Code Review Culture:** Every pull request receives a review from a Core Maintainer within 48 hours. This feedback loop — even for rejected PRs — is the primary teaching mechanism. Contributors learn by having their code read and critiqued by experienced peers.
4. **External Recognition Pipeline:** Active participation in programmes like Google Summer of Code, FOSS United events, and regional hackathons creates visibility. Each GSoC selection, each external contribution, each event participation attracts new contributors and validates the lab's quality to the Institute.

Why a Lab — Not a Club

VOSS Labs is intentionally structured as a laboratory, not a community or club. The difference is fundamental. A community hosts events, runs workshops, and optimises for attendance. A lab builds products, ships code, and optimises for contribution. VOSS does not hold workshops for the sake of

workshops. It does not organise hackathons for the sake of competition. The lab gathers to write code that solves real problems — collaboratively, transparently, and in public.

What VOSS Labs Does

- **Collaborative coding sessions** where students work together on production code, debug issues, and ship features in real time.
- **Code review circles** where contributors learn by reading each other's work, giving and receiving structured feedback on pull requests.
- **Technical documentation sprints** to make projects accessible to newcomers — because software that cannot be understood cannot be maintained.
- **Open build sessions** where anyone can join and contribute to ongoing work, with Core Maintainers available for guidance and pair programming.

What VOSS Labs Does Not Do

- **Workshop-centric programming** — attendance at a workshop is not a credential; shipped code is.
- **Elected positions or hierarchical committees** — authority is earned through contribution, not campaigning.
- **Closed-door meetings or private decision-making** — all technical decisions are made in public, on GitHub, where any contributor can participate.
- **Activities that look good on posters but produce no shipped code** — every activity must result in a commit, a PR, or a release.

"Labs ship. Communities meet. We are a lab."

What We Need — The Ask

A phased request for institutional partnership, structured from zero-cost immediate needs to long-term sustainability.

Phase 1 — Immediate (Zero Cost) — Next Semester

- **Official recognition** as a student technical society, granting the lab the same status as existing student bodies on campus.
- **Faculty advisor** — one faculty member for mentorship and administrative liaison (estimated commitment: 2-4 hours per week). This role is guidance, not governance — the lab remains entirely student-run and merit-based.
- **Shared lab space** — access to an existing computer lab for 2-3 hours per week for collaborative coding sessions.
- **Permission to hold collaborative coding sessions** on campus, announced through official channels.

Phase 2 — As VOSS Grows

- **Dedicated lab corner** with development systems, if physical space permits — a consistent location where contributors can work together.
- **Infrastructure support** — if the lab scales significantly, VOSS may request support for hosting projects on campus infrastructure rather than external platforms. No financial commitment at this stage; external funding will be explored first.

Phase 3 — Long-term Sustainability

- **Board of Core Maintainers** — when the founding Lab Lead graduates, a board of Core Maintainers (selected by contribution quality and community trust) collectively continues lab direction.
- **Natural succession** — the college identifies its most technically skilled students through their public contribution records. No separate assessment is needed; the GitHub history is the evaluation.
- **Self-sustaining operations** — the lab becomes financially self-sustaining through external grants, industry partnerships, and alumni contributions.

External Funding Sources

VOSS Labs will actively pursue external funding to ensure zero cost to the Institute:

- **FOSS United** — grants up to Rs. 1 lakh for workshops, Rs. 25,000 for collaborative events, swags for core team members, and ongoing mentorship support.
- **GitHub Education** — free developer tools valued at over \$100,000, including CI/CD credits, cloud hosting, and design tools; plus training resources and event sponsorship.
- **Google Summer of Code** — \$1,500 to \$6,000 per selected student, with VOSS serving as an organising pipeline.

- **Industry partners** — FOSS United maintains partnerships with 9+ technology companies including Zerodha and Frappe; active student open source labs regularly attract direct industry sponsorship.

Roadmap & Timeline

A twelve-month operational plan from June 2026 through May 2027.

Period	Phase	Key Activities
Months 1-2 (Jun - Jul)	Setup	VOSS announced via Technical Representatives in year WhatsApp groups. Collaborative coding sessions begin mid-June. Good-first-issues curated across all repositories. CI/CD pipelines and development tooling set up for all projects.
Months 3-4 (Aug - Sep)	Contribution Drive	Mentor-mentee pairing established. First 10 merged pull requests from new contributors. Weekly code review circles begin. First internal demo of shipped features to faculty and students.
Months 5-6 (Oct - Nov)	Product Focus	Ship vboard Phase 1 with live data integration. Pilot VERP attendance module with one department. Begin open source awareness activities and Google Summer of Code preparation sessions.
Months 7-8 (Dec - Jan)	Scale	Participate in external events including FOSS Hack. Junior mentorship programme — senior contributors guide first-year students. Technical blog posts published. Apply for FOSS United support.
Months 9-10 (Feb - Mar)	Recognition	Google Summer of Code applications submitted (target: 5+). Open source showcase with live demos for the campus community. Year 2 planning begins with contributor input.
Months 11-12 (Apr - May)	Evaluation	Measure all KPIs against Year 1 targets. Collect student and faculty feedback. Document Year 1 learnings, publish annual report. Establish alumni network for graduated contributors.

Note: Technical Representatives (TRs) from each academic year are invited to participate and form teams for college-facing projects within VOSS. All activities are focused on production contribution — no ceremonial events.

Key Performance Indicators

Measurable targets for Year 1 of VOSS Labs operations.

Key Performance Indicator	Year 1 Target
Registered active members	100+
Students with merged pull requests	50+
Collaborative coding sessions held	15+
Google Summer of Code applications	5+
Open source projects maintained	4+
GitHub stars (all projects combined)	100+
Technical blog posts published	10+
External event participations	3+

Risk Mitigation

Risk	Mitigation Strategy	Contingency
Low initial contributor turnout	TR network, good-first-issues, code review culture	Partner with existing student bodies for joint outreach
Faculty advisor unavailability	Clear 2-4 hour/week commitment defined upfront	Rotate between interested faculty members by semester
Project scope creep	MVP-first approach, defined release cycles	Deprioritise features, ship core functionality first
Lab Lead graduation	Board of Core Maintainers succession model	Transition planning begins 6 months before graduation

Team Structure & Governance

A flat, contribution-based meritocracy. No elections, no fixed positions, no gatekeeping.

VOSS Labs operates on a flat, contribution-based meritocracy. No elections, no fixed positions, no gatekeeping. Authority is earned through contribution quality and community trust. The lab rejects the traditional student-club structure of presidents and committees because that model creates politics without producing better software. At VOSS, the only status that matters is the quality of code you ship and the judgment you demonstrate in reviewing others' work.

Lab Lead

Harshal More (founding Lab Lead) sets the technical direction of the lab, holds repository administrative access, represents the lab in external communications, and serves as the primary liaison with Institute administration. The Lab Lead does not have unilateral technical decision-making power — those decisions are made by consensus among Core Maintainers. When the Lab Lead graduates, the Board of Core Maintainers collectively selects the next Lab Lead from among themselves based on contribution history, technical judgment, and demonstrated trust.

Core Maintainers

Core Maintainers are trusted contributors with merge access to all repositories. They review pull requests, approve changes, manage releases, and mentor newer contributors. Core Maintainers are selected by the Lab Lead based on sustained high-quality contributions and demonstrated technical judgment. There is no fixed number — the group grows with the lab. A contributor who consistently ships quality code and gives thoughtful reviews will naturally be invited to join.

Contributors

Anyone who has had a pull request merged into any VOSS Labs project is a contributor. This is the only membership category — it requires no invitation, no approval, and no formality beyond code quality. Contributors can propose features, report bugs, write documentation, and submit pull requests. A contributor today can be a Core Maintainer tomorrow. The pathway is open and entirely merit-based.

Succession Model

The VOSS succession model is drawn from labs that have operated successfully for decades:

- MIT SIPB elects keyholders through nomination by active members after sustained participation and demonstrated responsibility.
- RCOS at RPI transitions leadership through coordinators chosen by the existing coordinator group based on contribution and commitment.

VOSS Labs follows a similar merit-based model: the Board of Core Maintainers collectively decides on the next Lab Lead when a transition is needed. No elections, no campaigning — the person's code, reviews, and judgment speak for them. This model ensures continuity without bureaucracy. The lab's most committed and capable contributors naturally rise to leadership, and the transition process is transparent and predictable.

Closing

VOSS Labs represents a different model of student organisation — one built on craft, public accountability, and the belief that students can build software as good as anyone else. There is no secret sauce. The projects are public, the code is open, and the process is transparent. Anyone can verify the

quality of work by visiting the GitHub organisation. Anyone can see who contributes and how much. This is the point: in open source, there is nowhere to hide, and that is exactly why employers trust it.

The request is simple: recognise the lab, assign one faculty advisor, and allow occasional use of a lab space. In return, the Institute gains accreditation evidence, innovation metrics, zero-cost infrastructure, and students who enter the job market with verifiable, public proof of their abilities. The lab asks for no budget, no permanent space, and no special privileges. It asks only for the opportunity to do what it is already doing — at scale, with structure, and in partnership with the Institute.

"Your GitHub contribution history is your membership card."

Respectfully submitted,

VOSS Labs Founding Team
Vidyalankar Open Source Software Labs
Vidyalankar Institute of Technology, Mumbai
May 2026

Learn more: Website: vosslabs.org | Code: github.com/voss-labs

About the Lab Lead

Harshal More — Lab Lead, VOSS Labs

Harshal More serves as the Lab Lead for VOSS Labs. He is a founding member of the laboratory and oversees its technical direction, production standards, and external partnerships. His role is to ensure that every project shipped under the VOSS banner meets the quality bar of real-world, production-grade software — the same standard that industry demands.

Background

Harshal is a third-year undergraduate in Electronics & Computer Science at Vidyalankar Institute of Technology (CGPA: 8.1). His experience spans production AI infrastructure and widely-adopted open source projects:

- **Memori Labs** — Creator & Lead Architect of Memori, a SQL-native memory layer for LLMs. Grew to 12,000+ GitHub stars and 60,000+ SDK downloads.
- **Swarms** — Core Contributor to an enterprise multi-agent framework (5,600+ stars). Built test infrastructure and authored production architecture guides.
- **OpenAI gpt-oss** — Contributed a Gradio chat interface with function-calling to OpenAI's official open source project.

- **Naval-Gemma** — Fine-tuned Google's Gemma-3B for philosophical Q&A, deployed via Ollama for private inference.

Why This Matters for VOSS

Harshal has shipped production systems used by thousands of developers worldwide. He brings that same standard — clean architecture, comprehensive testing, clear documentation — to VOSS Labs. The goal is not student projects that collect dust. The goal is software the college uses every day, that other colleges can adopt, and that every contributor can point to as proof of their capability.

The Commitment

Every skill Harshal has learned in industry — from architecting multi-database systems to writing documentation that actually helps — is what he aims to share with every VOSS contributor. The lab grows when its contributors grow. The college's name gets bigger when its students ship software the world notices. VOSS is not about one person's profile. It is about creating an environment where every contributor can build a profile equally worth noticing.

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